Tilt Maze – ET095G Project Report

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1. Introduction

This report describes the design and implementation of the embedded game *Tilt Maze*, developed as part of the ET095G course. The purpose of the project was to build a complete interactive system that integrates multiple hardware components previously used in the course, including sensors, analog inputs, timers, LCD graphics, LEDs, and PWM-generated sound.

The game challenges the player to tilt the device in order to move a small ball on the LCD screen. The objective is to reach a randomly placed goal within a 10-second time limit. The game consists of five rounds, each with increasing movement speed and difficulty. A potentiometer is used to adjust the difficulty dynamically by scaling the movement speed. A start button initiates the game, sound effects are generated through PWM, and LEDs are used for visual celebration on the final win screen.

The project demonstrates real-time control, sensor-driven input, graphics rendering, time management, and efficient use of embedded resources.

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2. Implementation Analysis

The system integrates several hardware components working together. Each major subsystem is described below.

2.1 Accelerometer (MMA7660)

The MMA7660 accelerometer is used to detect tilt in the x- and y-axes. It communicates through the I2C interface and is configured for a 32 Hz sampling rate.
Only x and y readings are used to determine the direction and speed of the ball.

Tilt values are scaled and multiplied by both the round difficulty and the potentiometer difficulty. This provides smooth and intuitive control where the player simply tilts the device to move the ball.

2.2 Analog Input (Potentiometer – ADC)

A potentiometer connected to pin p19 is used as an analog input. The ADC reading ranges between 0.0 and 1.0 and is used to adjust the difficulty.
The value directly affects a scaling factor that modifies the ball's speed, resulting in a multiplier between 1.0 and 3.0.

This gives the player full control over how fast or challenging the game should be.

2.3 LCD Display (C12832)

The LCD screen shows all game graphics and text:

- Ball position
- Goal position
- Timer
- Round number
- Score
- Start menu
- Round win screen
- Final victory screen

Simple shapes (circles and filled circles) are used to render the game objects. The screen is cleared and redrawn regularly to maintain smooth animation.

2.4 Timer

A timer is used to manage the 10-second time limit for each round.
When the timer exceeds 10 seconds, the round is lost and a “Game Over” screen is shown.

The timer is also used to calculate the score. The remaining time (10 − elapsed time) is multiplied by 10 and added to the player’s total score when the goal is reached.

2.5 Speaker (PWM)

The game uses a PWM output on pin p26 to generate simple sound effects.
Different tones are produced by adjusting the PWM period. Sound effects include:

- A short tone when the game starts
- A rising sequence when a round is completed
- A descending “loss sound” when the timer expires

These sound effects improve the user experience and give immediate feedback.

2.6 LEDs

Four onboard LEDs are used to create a blinking animation at the end of the fifth round.
They blink several times before the final victory sound plays.

This provides a simple visual reward to the player after completing the game.

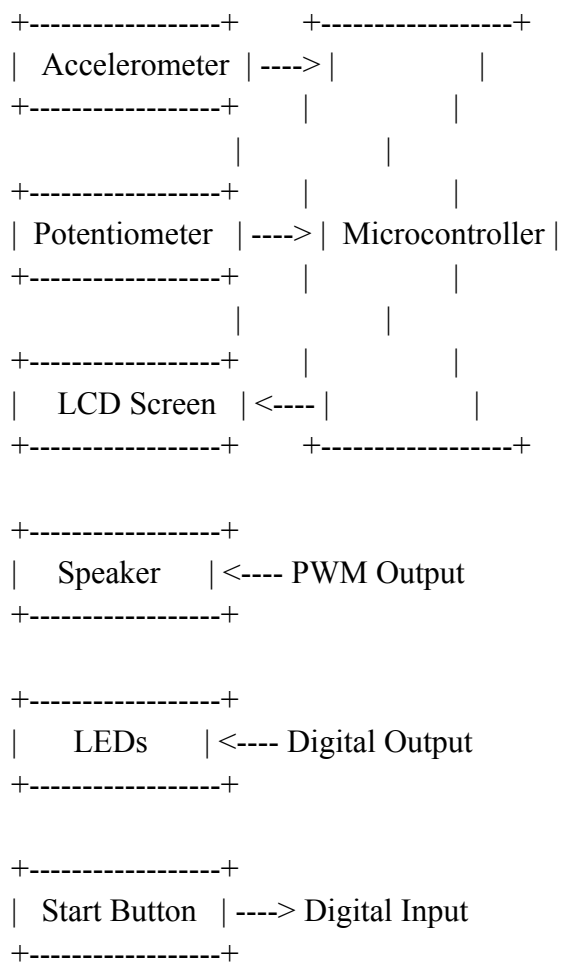
2.7 Start Button

A digital input button is used to start the game.
A short delay (debouncing) prevents the button from triggering multiple times in the same press.

The game does not begin until the player presses this button, ensuring proper control flow.

2.8 System Block Diagram

Below is a clear block diagram illustrating how the system components are connected conceptually:



3. Discussion

The project successfully integrates multiple hardware features into a coherent, real-time embedded game. The accelerometer provides smooth and intuitive control, while the potentiometer allows adjustable difficulty. Visual and sound effects enhance the gameplay experience.

Performance and Constraints

The main performance limitation is LCD redraw speed. Since clearing and redrawing the display takes time, the system uses a controlled update frequency to avoid unnecessary overhead.
The accelerometer data contains some natural noise, but within this game context it does not create major problems.

Potential Improvements

If expanded into a real consumer product, the following improvements would be beneficial:

- Adding filtering to smooth accelerometer noise
- Improving button debouncing with interrupts
- Power-saving features for battery operation
- More advanced graphics and animations
- Expanded sound effects
- Pause functionality

Overall, the implementation meets all course requirements and provides a complete, functional embedded game demonstrating several key concepts.

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4. Conclusion

Tilt Maze successfully demonstrates the use of multiple embedded components in a coordinated interactive application. The project highlights key concepts such as sensor integration, analog input, timer-based control, LCD graphics, PWM audio, and digital IO.
The result is a fully playable game that includes increasing difficulty, scoring, sound effects, and a complete user experience from start menu to final win screen.